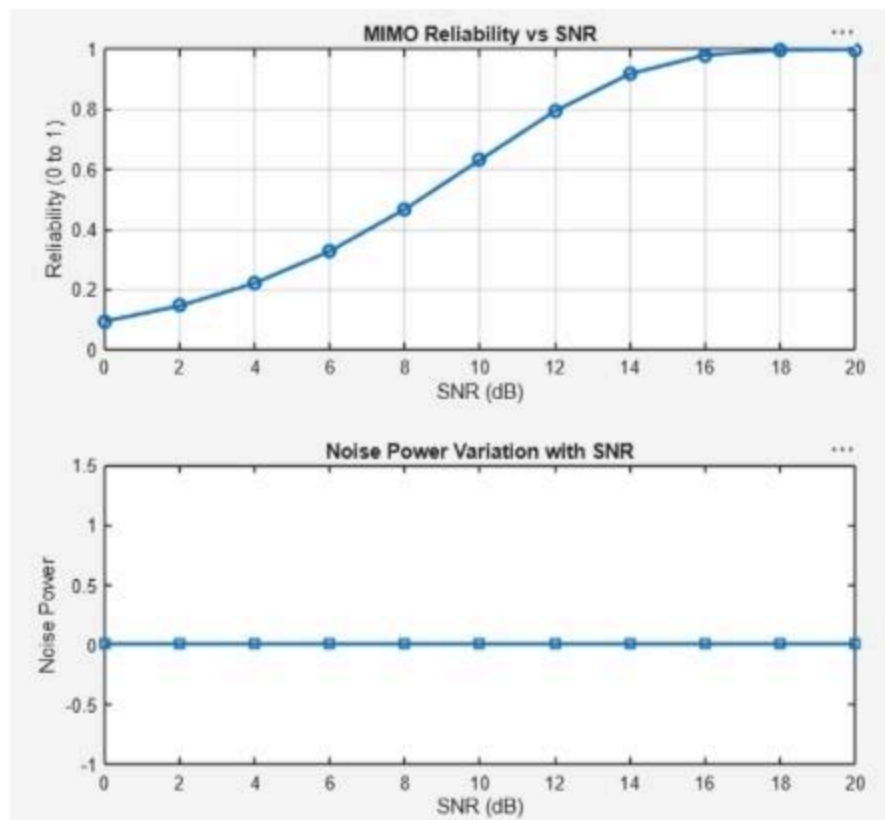


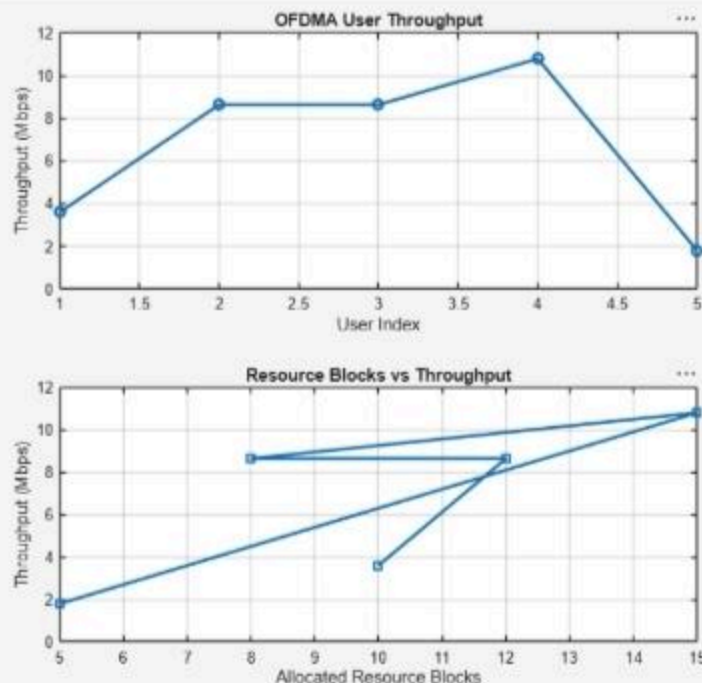
Objective 2: Matlab Code and Output

```
33 % SNR range (dB)
34 SNR_dB = 0:2:20;
35 SNR = 10.^(SNR_dB/10);
36 % MIMO parameters
37 Nt = 2; Nr = 2;
38 % Simulated reliability metric (inverse BER model)
39 Reliability = zeros(1,length(SNR));
40 for i = 1:length(SNR)
41     % simple channel model effect
42     H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
43     noise_power = 1/SNR(i);
44     % reliability increases with SNR
45     Reliability(i) = 1 - exp(-SNR(i)/10);
46 end
47 figure
48 % ----- Graph 1: SNR vs Reliability -----
49 subplot(2,1,1)
50 plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
51 title('MIMO Reliability vs SNR')
52 xlabel('SNR (dB)')
53 ylabel('Reliability (0 to 1)')
54 grid on
55 % ----- Graph 2: SNR vs Noise Effect -----
56 subplot(2,1,2)
57 plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
58 title('Noise Power Variation with SNR')
59 xlabel('SNR (dB)')
60 ylabel('Noise Power')
```



Objective 3: Matlab Code and Output

```
62 % Resource Block parameters
63 rb_bw = 180e3; % 180 kHz per RB
64 % Users
65 users = 1:5;
66 % Allocated Resource Blocks per user
67 alloc_RB = [10 12 8 15 5];
68 % Modulation efficiency (bits/symbol)
69 mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
70 % Throughput calculation (Mbps)
71 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
72 % Display results
73 disp('User Throughput (Mbps):')
74 disp(throughput')
75 figure
76 % ----- Graph 1: Throughput per User -----
77 subplot(2,1,1)
78 plot(users, throughput, '-o','LineWidth',2)
79 title('OFDMA User Throughput')
80 xlabel('User Index')
81 ylabel('Throughput (Mbps)')
82 grid on
83 % ----- Graph 2: RB vs Throughput -----
84 subplot(2,1,2)
85 plot(alloc_RB, throughput, '-s','LineWidth',2)
86 title('Resource Blocks vs Throughput')
87 xlabel('Allocated Resource Blocks')
88 ylabel('Throughput (Mbps)')
89 grid on
```



Ex.No.5: MIMO System Simulation with BER Analysis Date

Objective 1: Matlab Code and Output

/MATLAB Drive/slot5.m

```
1  clc; clear; close all;
2  % SNR range
3  SNR_dB = 0:2:20;
4  SNR = 10.^(SNR_dB/10);
5  % MIMO parameters
6  Nt = 2; Nr = 2;
7  BER = zeros(1,length(SNR));
8  for i = 1:length(SNR)
9      bits = randi([0 1],10000,1);
10     symbols = 2*bits-1; % BPSK
11     H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
12     noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
13     tx = H(1,1)*symbols + noise(1,:);
14     rx = real(tx) > 0;
15     BER(i) = sum(rx ~= bits)/length(bits);
16 end
17 figure
18 % ----- Graph 1: BER vs SNR -----
19 subplot(2,1,1)
20 semilogy(SNR_dB,BER,'-o','linewidth',2)
21 title('MIMO BER Performance vs SNR')
22 xlabel('SNR (dB)')
23 ylabel('Bit Error Rate (BER)')
24 grid on
25 % ----- Graph 2: SNR vs Quality Indicator -----
26 subplot(2,1,2)
27 plot(SNR_dB,1-BER,'-s','linewidth',2)
28 title('MIMO System Reliability vs SNR')
29 xlabel('SNR (dB)')
30 ylabel('Reliability (1 - BER)')
31 grid on
32
```

